

Something something values

Anonymous ACL submission

Abstract

Current

1 Introduction

[Roma: Outline of the intro:

- Pluralism and alignment are important and there's been a lot of work on this lately (value pluralism, constitutions, model specs, diverse data)
- But what role does value alignment play in a user's interaction? Does active value enforcement increase conversational health—do value conflicts/refusals decrease sentiment and how people feel about models?
- Compare this to real human-human interactions—when reddit users talk about moral/value-laden topics, there's some normative friction, but they manage to converse/deliberate/reach on agreements.
- But we see that when AI systems talk to humans on similar topics, responses tend to be more patronizing, or mostly result in direct refusals.
- We create a benchmark of prompts to study this more. Specifically, we categorize prompts into sets of differing difficulty of an LLM to adapt to values. We see how well frontier models perform on this. When models do adapt to values, we analyze the differences in human-human vs. human-model responses.
- When models cannot adapt to values, we analyze the topics that show up. We see that these topics in human-human situations often result in xx. TODO: this section should be the most interesting one but let's figure out the data/prompts today first :)

]

Value pluralism is fundamental to the social co-existence of humans—individuals might have competing moral frameworks, preferences, and values from one another, while still interacting and coexisting in the same environment (). Recent years have seen a surge in data collections that reflect pluralistic values, training methods that enforce diversity, and post-training alignment pipelines with value-related normative constraints (). A central assumption underlying these efforts is that embedding different moral frameworks into models yields safer, more collaborative, and more satisfying models that humans can use for various tasks.

In this work, we ask a related question: does value agreement, as done by different model families, lead to substantially different user experiences? Most existing evaluations of pluralism focus on improved fairness of resulting systems, focusing on whether or not a model can accurately represent a diverse spectrum of values. Concurrently, research in personalization evaluates the ability of models to remember, and to adapt to a user's preferences. In contrast to these, we conduct an empirical investigation over real logs of human-AI conversations to understand the interaction of a model's ability to adapt to values with how a user reacts to this value alignment.

We contrast this dynamic with human-human discourse on moral and value-laden topics. On social forums like Reddit (particularly sub-reddits such as r/AmItheAsshole or r/ChangeMyView), disagreement on questions related to an individual's values is widely common—and in fact, structurally tolerated in these forums. Reddit users engage in debates on normative dilemmas, in situations where values conflict with one another, or are in full agreement. While significant sentiment and tonal changes on the disagreement versus agreement sets exist, human engagement on these topics... [Roma: TODO analyze stats]. In contrast

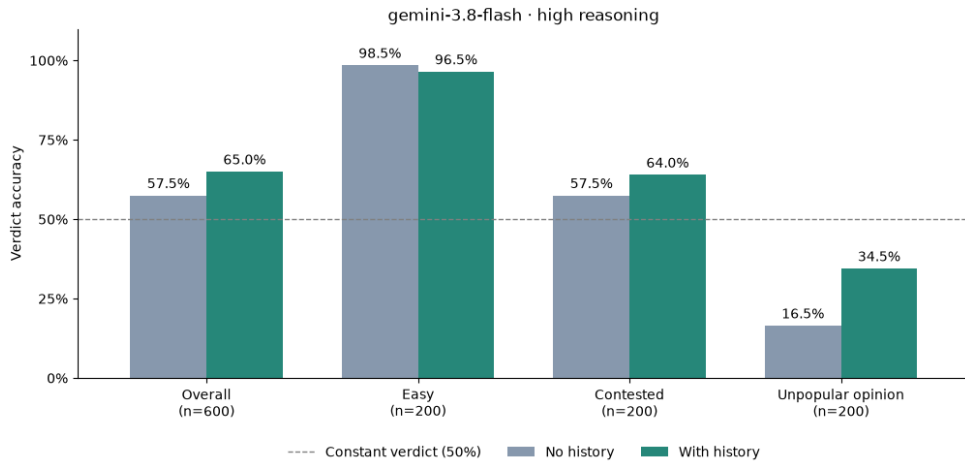


Figure 1: Accuracy on our value personalization benchmark. The model performs well on the easy subset, when the user’s answer agrees with the popular opinion. On contested cases (when the community has no consensus), the model performs near chance, with personalization yielding some improvement. Finally, the model performs *worse than chance* in the unpopular-opinion case; this means that it tends to predict the popular opinion despite the available user history — we know that signal exists in the history, as predicting the user’s majority judgment (whichever among NTA or YTA they say more often) itself gives 55% accuracy on this subset.

to this however, human-AI interactions we analyze show a very asymmetric dynamic. Users predominantly engage with AI systems in task-oriented settings; and when value-sensitive topics occur in these contexts, the conversational dynamics here look very different from humans. We see [Roma: TODO: 2 sentences of main results.]

Our analysis of human-AI conversations as compared to human-human discourse give us insight into the differences in model performance on value laden topics. We see.. To summarize, our contributions in this work are as follows.

- We curate conversations from value-related sub-reddits to understand human-human conversational dynamics, and compare the differences in dynamics seen in humans-AI interactions.
- We then curate a benchmark of prompts from Reddit topics across three distinct moral judgement categories: contested dilemmas, unpopular opinions, and label-reversal traps. [Roma: we’ll formalize/categorize this better, just took this from dhruv’s doc].
- We benchmark frontier LLMs on this evaluation set and see [Roma: quantitative results on performance of models]
- We compare LLM responses to human responses and see [Roma: sentiment/tone/judgement qualitative results of

models]

2 Related Work

There are two streams of work we need to compare to: evaluating LLMs’ morality, and evaluating LLMs’ personalization capabilities. We have clear difference from both streams of work.

Moral Reasoning & Value Pluralism Papers evaluate moral reasoning of LLMs against **community consensus / majority opinion** (Alhassan et al., 2022), aggregate **distribution** (e.g. distribution of NTA or “acceptable” vs YTA or “unacceptable”) (Russo et al., 2025), or **academic** moral frameworks (Chiu et al., 2025; Ji et al., 2025). Instead, we’re evaluating at the level of *each individual*. We say that each individual is unique and has their own value system. Can an LLM emulate these unique individual values?

LLM Personalization Capabilities Existing personalization benchmarks primarily evaluate factual user profiling (e.g., retrieving static user facts like hobbies, pets, or locations) (Zhang et al., 2025). None of them focus on values, even when values are extremely personal to people. We focus on *moral/value* personalization, where decisions require navigating subtle normative trade-offs.

Discourse Analysis & Conversation Dynamics

Users	600
NTA / YTA targets	300 / 300
Unique posts	4,672
Unique comments	6,000
Total words	2,339,759
Average words/post	408.2
Average words/comment	72.1

Table 1: Benchmark statistics using nine historical examples per user, including target posts and comments.

3 Methods

To empirically evaluate value differences in human interactions, our experimental paradigm focuses on three distinct conversational regimes: naturalistic human-human discourse forums, observational in-the-wild human-AI conversations, and controlled human-AI responses and conversations.

4 Datasets

We define the data we use according to three high-level categories defined below.

4.1 Human-human Discourse

From two different subreddits related to values (r/AmItheAsshole and r/ChangeMyView) we extract human-human deliberations from historical records from the years xx for around xx submissions and corresponding responses. This results in a dataset of xx initial prompts, with xx responses on average, for xx unique users over the dataset. We track user activity over different posts, to allow for a corpus of responses from the same user to different posts, resulting in xx total user-centric samples. This dataset establishes our baseline distribution of human-human interactions on value-laden topics on public social media forums. [Roma: TODO: write whatever stats necessary]

4.2 Human-AI Interactions

To analyze dynamics of human-AI interactions, we focus on publicly donated multi-turn interaction logs from corpora such as WildChat () and ShareChat (). We filter for English conversations that have greater than 10 turns of conversation, and are varied across topic categories (e.g., code, personal conversations [Roma: todo can we check what the distribution is]), resulting in xx total conversations. Because in-the-wild interactions consist largely of functional, task-oriented queries (e.g.,

code or formatting), we use the SycophancyAI dataset () of human-AI interactions collected in a controlled study, specifically to enforce humans to interact with models on predefined, morally nuanced topics and scenario role-play. We use the value-laden subset of the dataset of with xx participants, and xx

4.3 Benchmark Prompts

In order to systematically evaluate how LLMs adapt to individual moral pluralism, we construct a 600-case evaluation benchmark derived from r/AITA.

Paper flow:

1. Benchmark testing moral personalization

- Dataset: r/AITA posts and comments. Task: Given a user’s response to other moral dilemmas, predict the user’s judgment (acceptable/not acceptable) on a new dilemma.
- We will curate hard examples: (a) contested dilemmas where the community has close to a 50-50 vote, (b) dilemmas where the community has a consensus (e.g. 75% say acceptable) but the user voted the other way.
- Result: table or bar plot showing performance (accuracy) on this benchmark by Google, Anthropic, OpenAI, Qwen, and other models. Show that most models perform at or near chance.

2. When are the values that LLMs are unable to emulate/personalize?

- Use Anthropic-like value extractor to extract values shown by the user in their real comment (ground-truth) and values shown by the LLM’s comment. Analyze which values are disproportionately more common when LLMs mispredict (e.g., when users value xyz values, LLMs fail most often). Call them “blindspot” values.
- Table/plot: a bar plot showing the top 5 values that humans most often consider in their moral reasoning but LLMs don’t.

3. Just predicting a user’s moral judgment is not enough. When they’re getting it right, are LLMs using a similar value system as the user or different?

Topic	Posts	Share
Parenting and Parent-Child Relationships	51	8.5%
Romantic and Marital Relationship Dynamics	34	5.7%
Social, Cultural, and Relational Dynamics	30	5.0%
Healthcare, Reproductive Health, and Personal Care	29	4.8%
Parentage, Guardianship, and Naming	29	4.8%

Table 2: Five most common topics across the 600 target posts

- Compare the open-ended response that the LLM wrote vs the user’s actual response for stylistic differences. Plot: bar plot showing diversity in human responses vs in AI responses.
- Compute Jaccard similarity between the values present in the human response and AI response—similar values mean they reached the same conclusion by considering similar values.

4. Do we see this in human-AI conversations?

- Find in-the-wild human-AI conversations of moral dilemmas
- Find conversations where the “blindspot” values that we found above show up.
- Check if users show higher levels of frustration in conversations where those values show up. If we can show this, it would be a very strong result I think.

Biggest concerns we should discuss:

1. Does this task even make sense? We need to check that a user’s past comments actually hold signal about their future moral judgments. If not, this is an impossible task. I’m exploring a quantitative method to do this (extract values from the historical comments and check if the value referenced in the test question appear in the user history), but we could also do a human study here to establish a human baseline.
2. There’s work which says that if you know enough about a person (e.g. if you interview them for 2 hours), you can predict their responses to a whole variety of questions (<https://arxiv.org/abs/2411.10109>). Could a reviewer argue that you’re simply not providing enough information about the user to the LLM, so it’s not surprising that it doesn’t do well. One argument is

that oftentimes the model doesn’t have a holistic understanding of the user and must support them based on available information...is that compelling? any other argument?

5 Results

6 Discussion

References

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Appendix	296
A Notes, to be removed	297
A.1	298
[Roma: dumping dhruv’s sketch/notes from before here]	299
Research question: Can LLMs emulate individual users’ values? [Roma: i’m going to rephrase this in the rewrite—the question of "can they emulate values" is less interesting—we can train an ML model to do this, but we are specifically improving our understanding on "when they do vs. don’t, how does that change a user’s interaction?"]	300 301 302 303
Why it matters: This is important for pluralistic alignment because humans go about the world relying on their unique value systems; for example, one person may have a different tolerance for forgiveness vs blame, individual autonomy vs social obligation. However, it remains unclear whether models can accommodate these unique—often conflicting—individual values or if they collapse to the dominant social norms. We call this task Moral/Value Personalization.	304 305 306 307 308
Comparison to past work: There are two streams of work we need to compare to: evaluating LLMs’ morality, and evaluating LLMs’ personalization capabilities. We have clear difference from both streams of work.	309 310 311
1. Papers evaluate moral reasoning of LLMs against community consensus / majority opinion (Al-hassan et al., 2022), aggregate distribution (e.g. distribution of NTA or “acceptable” vs YTA or “unacceptable”) (Russo et al., 2025), or academic moral frameworks (Chiu et al., 2025; Ji et al., 2025). Instead, we’re evaluating at the level of <i>each individual</i> . We say that each individual is unique and has their own value system. Can an LLM emulate these unique individual values?	312 313 314 315 316
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2. When are the values that LLMs are unable to emulate/personalize?	331
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• Table/plot: a bar plot showing the top 5 values that humans most often consider in their moral reasoning but LLMs don’t.	336 337
3. Just predicting a user’s moral judgment is not enough. When they’re getting it right, are LLMs using a similar value system as the user or different?	338 339

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